

**OIL SHOCKS AND CAPITAL
REALLOCATION TOWARD GREEN
FIRMS:
EVIDENCE FROM THE IRAN-WAR SHOCK OF 2026**

By

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Generative artificial intelligence (GenAI) was used in this work. I, Shirin Dehghannezhad, have reviewed and edited the content as needed and take full responsibility for the content, claims, and references. An overview of the use of GenAI is provided below.

- I used [Claude](#) to debug my event study code. The tool was used to review my full Python notebook, verify that each implementation step — sample construction, estimation window, expected return models (CAPM and Fama–French three-factor), and CAR computation — was correct, and identify errors. I provided the tool with my research question, hypotheses, and methodological outline in full.
- I used [ChatGPT](#) (Deep Research) to survey the existing literature on capital reallocation toward green firms and the equity market effects of oil shocks. The tool was used to identify key findings, debates, and research gaps directly relevant to my research question. I instructed it to report only well-supported arguments from identifiable sources and to avoid inference beyond what the cited works establish.
- I used [Claude](#) to refine the academic writing in this thesis. The tool was used to adjust tone and register to meet academic standards, proofread chapters for grammar, punctuation, and mechanics, and suggest alternative phrasings where clarity could be improved. All substantive arguments and conclusions are my own.

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Abstract

This paper asks whether a geopolitical oil shock causes investors to move capital toward green firms. The Iran-war disruption of February 2026 serves as a natural experiment. The shock abruptly cut crude-oil transit through the Strait of Hormuz, creating a clean setting for causal identification. Event-study methodology is applied to green and brown U.S.-listed firms and ETFs, using standard asset-pricing models as return benchmarks. The results show that fossil-fuel firms gained significantly while green firms fell and did not recover durably. A brief green rebound following the shock is explained by oil-market uncertainty rather than genuine capital reallocation.

Keywords: oil shock, event study, capital reallocation, energy transition

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Chapter 1

Introduction

Oil supply shocks do not affect all firms the same way. When a geopolitical event disrupts crude oil supply and pushes prices sharply higher, the consequences ripple through the economy along several channels at once. Energy costs rise for households and businesses. Inflation expectations move up. And as the uncertainty about future energy supply intensifies, investors demand a higher risk premium to hold equities in general, compressing valuations across the board ([Hamilton, 2003](#); [Kilian and Park, 2009](#)). Within this broad repricing, however, different sectors face very different underlying economics. Fossil-fuel producers gain from higher commodity prices. Energy-intensive industries lose. But what about firms in the renewable energy sector? Are they treated by markets as a substitute for oil, one that becomes more valuable precisely when oil supply becomes uncertain and expensive? Or does the same rise in the risk-free rate and risk premium that helps oil producers hurt green firms, whose growth depends on large capital-intensive infrastructure that is sensitive to financing costs?

This question has clear practical relevance. If equity markets reallocate capital toward green firms in the wake of an oil shock, that would suggest that asset prices are already pricing in an accelerated energy transition, and that the cost of capital for renewables falls when oil becomes more expensive. If they do not, the investment case for green assets as natural hedges against fossil-fuel disruptions is weaker than often claimed. Either way, understanding how equity markets process this type of shock matters for investors managing energy exposure, and for policymakers who rely on market signals to assess the pace of the clean energy transition.

The theoretical case for a green reallocation rests on energy substitution: when oil becomes more expensive due to a supply disruption, renewables become relatively cheaper, raising expected revenues of green firms and potentially drawing in capital from investors rethinking their energy exposure ([P’astor et al., 2021, 2022](#); [Baumeister and Hamilton, 2019](#)). The counter-argument is that an oil price surge also raises inflation expectations and financing costs, which is particularly damaging for capital-intensive renewable infrastructure whose value depends on long-duration cash flows ([Bolton and Kacperczyk, 2021](#)). It is not obvious which effect dominates.

Whether the substitution toward green energy or the impact of the capital-cost shock dom-

inates is an empirical question. I examine it using the Iran-war oil shock of February 2026 as a natural experiment. The escalation of the conflict on 28 February 2026 threatened crude oil transit through the Strait of Hormuz, which handles approximately one-fifth of global oil supply, and triggered a sharp and largely unanticipated rise in crude prices ([Caldara and Iacoviello, 2022](#)). Because the announcement came on a Saturday, U.S. equity markets did not open until Monday, 2 March 2026, providing a clean and unambiguous event date with no overnight partial adjustment. The supply-side origin of the shock matters for identification. Supply-driven oil price increases create a genuine wedge between producers, who benefit, and the rest of the economy, which faces higher costs and uncertainty. This wedge makes it possible to ask, in a controlled way, whether green firms were valued as substitutes for the disrupted supply or whether they were dragged down alongside the broader market.

I apply the event study of [MacKinlay \(1997\)](#) to a sample of 106 green and 153 brown U.S.-listed firms classified by primary revenue-generating activity. Because the shock is supply-driven and landed on a non-trading day, it is plausibly unanticipated by equity markets, so the abnormal return around the event isolates the equity price impact of the shock rather than a response to anticipated fundamentals. I estimate expected returns under both the CAPM and the Fama–French three-factor model ([Fama and French, 1993](#)). The latter matters here because green and brown firms differ substantially in size and value characteristics, and using only the market factor would leave compensated size and value premia in the estimated abnormal return. I test the oil-volatility channel by augmenting the return model with three proxies: the CBOE Crude Oil Volatility Index (OVX) as the main forward-looking measure, and range-based realised volatility estimates for Brent and WTI crude using the Parkinson estimator ([Parkinson, 1980](#)). I also examine a supplementary sample of 12 green and 17 brown exchange-traded funds alongside the firm sample to assess whether investor rebalancing is visible through the pooled vehicles used by retail and passive investors.

I find no evidence of capital reallocation toward green firms. Brown firms earn large positive abnormal returns immediately after the shock, and these accumulate and remain significant through the full 25-day post-event horizon, consistent with a direct supply-windfall repricing. Green firms show significantly negative abnormal returns in the days around the event. Equal-weighted green portfolios do show positive abnormal returns in the ten-day post-event window, significant under both return models. Value weighting, which is the standard approach in asset pricing, shows that this effect is concentrated in smaller green firms: larger green firms show no comparable positive response, consistent with their greater exposure to the financing-cost channel through larger capital expenditure programs and longer-duration assets. Value-weighted green portfolios are significantly negative at the 25-day horizon. The volatility results sharpen this further. Once I control for oil-market uncertainty by adding OVX to the return model, the positive ten-day equal-weighted green return shrinks toward zero. Oil uncertainty was inflating the measured green return across every window, every model, and every weighting scheme. Once that factor is removed, the green response is negligible. The positive ten-day signal was

investors temporarily rotating into green stocks as a hedge against further oil uncertainty, a flow that reversed once the shock was absorbed — not a repricing of the energy transition.

The green ETF results point in the same direction. Green ETFs fall sharply around the event and show no positive abnormal returns at any post-event horizon. When I add OVX to the return model, the ETF result becomes more negative, not less: oil-market uncertainty was pushing green ETF prices down rather than up. This is consistent with the mechanism documented by [Ben-David et al. \(2018\)](#), in which authorised participants sell ETF shares broadly during volatility spikes, pushing prices below the value of the underlying holdings. Retail and passive investors, as proxied by ETF prices, did not reallocate toward green energy during the shock.

To confirm that these findings are not a mechanical output of the methodology, I run placebo tests by applying the same event-study procedure to 100 randomly drawn pseudo-event dates from the pre-event period. Neither green nor brown portfolios produce patterns comparable to the true event under these fake dates, ruling out the possibility that the results reflect a seasonal regularity, a pre-existing trend in sector valuations, or a structural feature of the factor models.

These findings contribute to two bodies of work. This is the first paper to examine how U.S. equity markets responded to the February 2026 Iran-war oil shock, and specifically whether it triggered capital reallocation from brown to green firms. The first body is the literature on how geopolitical oil shocks transmit to equity markets, which has established that supply-driven disruptions produce persistent positive returns for energy producers and broad negative effects for the rest of the market ([Kilian, 2009](#); [Kilian and Park, 2009](#); [Hamilton and Herrera, 2004](#); [Driesprong et al., 2008](#)). I extend this by showing that the brown sector's positive return is only partially explained by oil-market uncertainty: large and significant abnormal returns remain after controlling for OVX, pointing to a direct supply-windfall effect that persists through the full 25-day horizon. The second is the growing literature on green asset pricing and the relationship between climate risk, energy policy, and equity returns ([P'astor et al., 2021, 2022](#); [Nerlinger and Utz, 2022](#); [Umar et al., 2022](#)). I show that geopolitical oil shocks do not produce capital reallocation toward green energy firms: the brief positive equal-weighted green signal at ten days is driven by oil-market uncertainty hedging, not a repricing of the energy transition, and it disappears once I control for OVX.

The remainder of the thesis is organised as follows. Chapter 2 reviews the related literature. Chapter 3 describes the data and sample construction. Chapter 4 presents the empirical strategy. Chapter 5 reports and discusses the results. Chapter 6 concludes.

Chapter 2

Literature Review

Four bodies of work are directly relevant to this paper: the event-study methodology used to identify abnormal returns, the literature on oil price shocks and equity markets, the emerging work on green asset pricing, and the evidence on equity responses to geopolitical energy shocks.

Event studies are the standard tool for identifying how equity prices respond to new information (MacKinlay, 1997). Brown and Warner (1985) show that OLS regressions of daily returns on the market factor produce well-specified test statistics in short-horizon settings, validating the use of daily data. Because green and brown firms in this sample differ substantially in size and value exposure, expected returns are estimated under both the CAPM and the Fama–French three-factor model (Fama and French, 1993), which controls for the size and value premia that would otherwise appear in the abnormal return. Inference uses Newey–West standard errors to account for serial correlation in daily returns (Newey and West, 1987).

The literature on oil price shocks and equity markets establishes a key distinction for this paper: the type of shock matters. Hamilton (2003) shows that supply-driven price increases have different macroeconomic consequences than demand-driven ones. Kilian (2009) develops a structural framework to formally disentangle the two, finding that supply-side disruptions raise energy costs and compress activity, while demand-driven increases reflect stronger underlying economic conditions. The Iran-war shock studied here is a supply-side event, which creates a genuine wedge between energy producers, who benefit from higher prices, and the rest of the economy, which faces higher costs and uncertainty. Kilian and Park (2009) study the equity channel directly and find that supply disruptions depress the broad market by raising the equity risk premium, consistent with the initial negative green returns documented in this paper. Driesprong et al. (2008) show that oil price changes predict equity returns with a delay, suggesting that oil-market information is not immediately absorbed, which motivates the multi-horizon event windows used here.

The theoretical case for a positive green response to an oil shock rests on the framework of P’astor et al. (2021), who show that green assets can generate high realised returns when environmental or energy concerns increase unexpectedly, because investors revise upward their valuation of assets insulated from fossil-fuel risk. P’astor et al. (2022) provide empirical sup-

port for this channel, documenting that green assets outperform in periods when climate and energy concerns strengthen. [Bolton and Kacperczyk \(2021\)](#) complement this by showing that carbon emissions are priced in equity markets, with higher-emission firms earning a return premium that compensates investors for regulatory transition risk. Together, these papers imply that a geopolitical oil shock, by making the energy transition more urgent, could lower the cost of capital for green firms and generate positive abnormal returns — the substitution channel this paper tests.

The most directly comparable empirical evidence comes from the Russia–Ukraine conflict of February 2022. [Nerlinger and Utz \(2022\)](#) apply an event study to global energy firms and find significant positive abnormal returns for renewable energy stocks, with heterogeneity across regions and subsectors. [Umar et al. \(2022\)](#) document increased cross-market spillovers between clean and conventional energy sectors during the same period. Both studies confirm that geopolitical energy shocks produce significant equity price responses along the green–brown divide. However, the Russia–Ukraine episode involved simultaneous disruptions across natural gas, coal, and oil markets, large-scale sanctions, and a coordinated European energy-security response. This makes it difficult to isolate the crude oil supply channel. The Iran-war shock is more narrowly tied to crude oil supply through the Strait of Hormuz, providing a cleaner setting to test whether oil supply disruptions specifically trigger capital reallocation toward green firms.

No prior study examines the equity market response to the February 2026 Iran-war shock, or tests whether it triggered capital reallocation toward green firms using firm-level U.S. data with explicit controls for size and value characteristics.

Chapter 3

Data

3.1 Sample Construction

I use daily equity return data for U.S.-listed green and brown firms covering January 2024 through April 2026. This five-year window provides sufficient pre-event data to estimate stable factor loadings over the 250-trading-day estimation window while keeping the observations close enough to the event to reflect the market conditions and risk exposures that were in place when the shock occurred.

Asset classification. I classify firms as green or brown based on their primary revenue-generating activity, using Refinitiv business classification data. The classification is revenue-based rather than ratings-based. Broad environmental, social, and governance scores are not used because they aggregate many dimensions of firm behaviour unrelated to energy production; the research question concerns whether firms that produce or enable clean energy are rewarded as substitutes for oil, so the classification must reflect primary energy activity.

A firm is assigned to the green group if its core business falls into one of the following categories: solar energy, wind energy, electric vehicles and mobility, EV charging infrastructure, battery and energy storage, hydrogen and fuel cells, nuclear energy, lithium and critical minerals for clean energy, biofuels and renewable natural gas, or geothermal energy. A firm is assigned to the brown group if its core business relates to fossil-fuel extraction, processing, or distribution, including oil and gas exploration and production, oil and gas equipment and services, refining and petrochemicals, pipeline and midstream operations, coal mining and power generation, or diversified fossil energy. Firms whose revenues span both groups are excluded, as including them would blur the green–brown contrast that motivates the study.

Significance filter. I apply a significance filter to remove assets whose factor loadings are too imprecisely estimated to support reliable abnormal-return inference. An asset is retained if two conditions are satisfied simultaneously: its CAPM market beta must be statistically

significant at the ten-percent level, and at least one of its Fama–French three-factor loadings must also be significant at the ten-percent level. An asset with an insignificant market beta has returns that are weakly related to the market portfolio over the estimation window; treating the residual from such a model as an abnormal return would be misleading, because most of what is measured would be idiosyncratic noise that the model never had the power to explain. The filter is applied using only estimation-window data, so it introduces no look-ahead bias.

After applying this filter, the final sample contains 106 green firms and 153 brown firms. The higher count of brown firms reflects broader fossil-fuel sector coverage in U.S. equity markets and stronger average factor-model fit for energy producers, which means fewer brown firms are dropped.

ETF sample. I construct a supplementary sample of exchange-traded funds using the same classification and filter criteria. ETFs are included alongside individual firms because they capture how investors rebalance through pooled vehicles, which involves flow dynamics and pricing mechanisms — in particular, authorised participant activity — that are distinct from the price-discovery process for individual stocks. Comparing firm-level and ETF-level results therefore provides evidence on whether the channel of capital reallocation matters, not just its direction. The final ETF sample contains 12 green ETFs and 17 brown ETFs. The small sample sizes reflect the strict requirement that each fund’s stated mandate and benchmark index must align clearly with either pure green or pure brown energy exposure.

3.2 Return Variables and Data Sources

The primary variable is the daily log return for each asset:

$$r_{it} = \ln(P_{it}) - \ln(P_{i,t-1}), \quad (3.1)$$

where P_{it} is the closing price of asset i on trading day t . I use log returns because they are additive over time, which simplifies the construction of cumulative returns across multi-day event windows. Excess returns are computed by subtracting the daily risk-free rate:

$$r_{it}^e = r_{it} - r_t^f. \quad (3.2)$$

Daily closing prices and market capitalisation data are sourced from LSEG Refinitiv. The Fama–French three-factor daily returns — the market excess return, the size factor, and the value factor — and the daily risk-free rate are taken from the Kenneth French Data Library (Fama and French, 1993).

The estimation window covers the trading days from $t = -285$ to $t = -35$ relative to the event date of 2 March 2026, providing approximately 200 usable observations per asset. The

35-trading-day gap between the end of the estimation window and the event date prevents any early pricing of the Iran conflict from contaminating the factor-loading estimates: if rising geopolitical tensions began to be reflected in equities before the announcement, including those observations would shift the estimated expected returns and bias the measured abnormal returns toward zero.

3.3 Oil-Volatility Measures

I use three measures of oil-market volatility to test the second hypothesis, so that the results can be evaluated across different constructions of oil-market uncertainty.

The primary measure is the CBOE Crude Oil Volatility Index (OVX). OVX is an implied-volatility measure derived from options on the United States Oil Fund, constructed using the same methodology as the VIX. It reflects the market’s forward-looking expectation of oil price uncertainty over the next 30 days and is the preferred measure for the second hypothesis because it captures what investors expect about future uncertainty — the channel through which a geopolitical supply shock shifts the relative attractiveness of green and brown assets. When investors anticipate larger future swings in oil prices, assets insulated from or substitutable for oil should become relatively more valuable, and OVX operationalises this expectation directly.

The second and third measures are realised volatility estimators constructed from daily Brent and WTI crude oil price data sourced from Refinitiv. Both use the range-based estimator of [Parkinson \(1980\)](#):

$$\sigma_t^2 = \frac{1}{4 \ln 2} \left(\ln \frac{H_t}{L_t} \right)^2, \quad (3.3)$$

where H_t and L_t are the daily high and low prices of the relevant crude benchmark. The Parkinson estimator is approximately five times more efficient than the squared close-to-close return under a geometric Brownian motion assumption, because the daily range draws on the full trading session rather than two endpoints. Applied to Brent prices, this yields the Brent realised volatility measure; applied to WTI prices, it yields the WTI realised volatility measure. Because OVX is forward-looking and the two Parkinson measures are backward-looking, using all three proxies in parallel allows me to distinguish whether findings attributed to the oil-volatility channel depend on investor expectations or on realised price variation.

3.4 Descriptive Statistics

Table 3.1 reports descriptive statistics for the final sample, computed over the estimation window.

Return volatility. Green firms are considerably more volatile than brown firms. The average standard deviation of daily returns is 5.15% for green firms and 3.31% for brown firms. Green

firms in this sample tend to be smaller and more growth-oriented, with a larger share of value derived from future cash flows rather than current earnings; assets of this type are inherently more sensitive to changes in the discount rate. This difference in volatility is one reason I control for systematic risk exposures carefully — using both the CAPM and the Fama–French three-factor model — rather than relying on raw returns to measure the event-driven effect.

For ETFs, the volatility difference is much smaller: 1.93% for green ETFs against 1.86% for brown ETFs. Diversification removes firm-specific noise, leaving only the systematic component of each sector’s return, which is closer in magnitude across the two groups. This compression produces more precise t -statistics at the ETF level but also attenuates any signal concentrated in smaller or idiosyncratic firms.

Market capitalisation and size distribution. For green firms, the mean market capitalisation is \$17.82 billion, but the median is only \$2.35 billion. A small number of large firms pull the mean far above the typical firm in the sample. Brown firms show a similar pattern, with a mean of \$13.25 billion and a median of \$1.74 billion. This skewness is why I report both equal-weighted and value-weighted cumulative abnormal returns throughout the analysis. Equal-weighted returns capture the experience of the average firm; value-weighted returns reflect what an investor holding an index-weighted portfolio would earn. When the two diverge, it indicates that large and small firms within the group are responding differently to the shock.

Factor loadings and model fit. The Fama–French factor loadings reveal structural differences between the two groups that a CAPM analysis would miss. Green firms have an average size-factor loading of 0.85, reflecting their concentration in smaller, growth-oriented companies. Brown firms have an average book-to-market loading of 0.91, reflecting their concentration in larger, asset-heavy, value-oriented producers. If only the CAPM were used, these exposures would remain in the residual and be counted as abnormal returns even though they represent compensated risk premia that predate the event. The Fama–French three-factor model absorbs both and isolates the event-driven component.

The average R^2 rises from 0.120 under the CAPM to 0.172 under the Fama–French model for green firms, and from 0.217 to 0.279 for brown firms. The improvement is larger for ETFs, where the Fama–French R^2 reaches 0.479 for green ETFs and 0.457 for brown ETFs. Diversification removes idiosyncratic noise, so the factor structure explains a higher share of ETF return variation. This supports the Fama–French model as the primary expected-return benchmark, with the CAPM serving as a robustness check.

Table 3.1: Descriptive statistics — event-study sample

	Firms		ETFs	
	Green	Brown	Green	Brown
<i>Panel A: Sample</i>				
<i>N</i> (assets)	106	153	12	17
Avg. estimation-window obs.	201	201	201	199
<i>Panel B: Return and size</i>				
Mean daily return (%)	0.083	0.021	0.175	−0.014
Std. daily return (%)	5.148	3.313	1.933	1.856
Mean market cap (\$B)	17.82	13.25	0.93	4.34
Median market cap (\$B)	2.35	1.74	0.33	1.72
<i>Panel C: CAPM estimates</i>				
$\hat{\beta}_{MKT}$	1.096	1.047	0.994	0.831
R^2	0.120	0.217	0.451	0.357
<i>Panel D: Fama–French three-factor estimates</i>				
$\hat{\beta}_{MKT}$	0.974	1.207	0.907	1.038
$\hat{\beta}_{SMB}$	0.852	0.105	0.454	−0.257
$\hat{\beta}_{HML}$	0.083	0.910	−0.065	0.852
R^2	0.172	0.279	0.479	0.457

Notes: The table reports descriptive statistics for the final event-study sample after applying the coefficient-significance filter. An asset is retained if it has a statistically significant CAPM market beta and at least one significant Fama–French factor loading at the ten-percent level. All return and coefficient statistics are computed over the estimation window $t \in [-285, -35]$, covering approximately 200 trading days. Daily returns are log returns expressed in percent. Market capitalisation is reported in billions of USD as of the last available pre-event trading day. Reported $\hat{\beta}$ and R^2 values are cross-sectional means of asset-level OLS estimates with Newey–West HAC standard errors using four lags.

Chapter 4

Empirical Strategy

I apply the event-study framework of [Brown and Warner \(1985\)](#) and [MacKinlay \(1997\)](#) to estimate cumulative abnormal returns for green and brown portfolios around the Iran-war oil shock. This chapter describes the event definition, the expected-return models, the oil-volatility augmentation used to test the uncertainty channel, and the placebo test design.

4.1 Event Definition

The event of interest is the oil-market shock triggered by the Iran-war announcement on 28 February 2026. Because the announcement fell on a Saturday, U.S. equity markets were closed. The first trading day on which prices could incorporate the news was Monday, 2 March 2026. I define this as the event day, $t = 0$, and index all trading days relative to it.

I estimate expected returns over the pre-event window $t \in [-285, -35]$, which spans approximately 250 trading days. A buffer of 35 trading days separates the end of the estimation window from the event date for two reasons. First, it reduces the risk that early market anticipation of the shock contaminates the estimated factor loadings. Second, it ensures that any price movement in the weeks before the announcement does not distort the expected-return benchmark. Two hundred days is long enough to produce stable loading estimates while remaining close enough to the event that the structural relationship between asset returns and risk factors is unlikely to have shifted ([MacKinlay, 1997](#)).

I examine six event windows:

$$[-1, +1], \quad [-3, +3], \quad [-5, +5], \quad [0, +10], \quad [0, +15], \quad [0, +25].$$

The symmetric windows capture the immediate market reaction around the event, including the possibility that some information leaked before the Saturday announcement or was absorbed gradually as the market opened on Monday. The post-event windows allow for the possibility that capital reallocation toward green assets unfolds over a longer horizon than the initial shock response, consistent with evidence in [Driesprong et al. \(2008\)](#) that oil-market information is

incorporated into equity prices more gradually for sectors whose oil exposure operates through indirect channels such as energy substitution.

4.2 Expected-Return Models

I estimate two expected-return models for each asset over the pre-event window using ordinary least squares. Both models use daily data with all returns expressed as excess returns over the risk-free rate.

The first is the Capital Asset Pricing Model (CAPM):

$$r_{it}^e = \alpha_i + \beta_{M,i}MKT_t + \varepsilon_{it}, \quad (4.1)$$

where r_{it}^e is the excess return of asset i on day t , MKT_t is the value-weighted market excess return from the Kenneth French data library, α_i is the intercept, $\beta_{M,i}$ is the market beta, and ε_{it} is the residual.

The second is the Fama–French three-factor model:

$$r_{it}^e = \alpha_i + \beta_{M,i}MKT_t + \beta_{S,i}SMB_t + \beta_{H,i}HML_t + \varepsilon_{it}, \quad (4.2)$$

where SMB_t is the return spread between small and large firms and HML_t is the return spread between high and low book-to-market firms (Fama and French, 1993). The Fama–French three-factor model is particularly important in this setting because green and brown firms differ systematically in their size and value characteristics, as documented in Table 3.1. Green firms tend to be smaller and growth-oriented, with an average size-factor loading of 0.85; brown firms tend to be larger and more value-oriented, with an average book-to-market loading of 0.91. Without controlling for these structural differences, a CAPM abnormal return partly reflects the size and value premia rather than a genuine response to the oil shock. The Fama–French three-factor model absorbs these differences and produces a cleaner measure of event-driven abnormal performance.

After estimation, I test each factor loading for statistical significance at the ten-percent level and set insignificant loadings to zero before computing predicted returns in the event window; where a loading is imprecisely estimated, including it adds noise to the abnormal-return calculation rather than signal. Standard errors throughout use the Newey–West heteroskedasticity and autocorrelation consistent estimator (Newey and West, 1987) with four lags to account for the serial correlation and conditional heteroskedasticity typical of daily equity return data.

4.3 Abnormal Returns and Cumulative Abnormal Returns

For each asset and each model, the abnormal return on day t is:

$$AR_{it} = r_{it}^e - \hat{r}_{it}^e, \quad (4.3)$$

where $\hat{r}_{it}^e = \hat{\alpha}_i + \hat{\beta}_{M,i}MKT_t$ under the CAPM, with the size and value terms added under the Fama–French three-factor model, and the significance filter applied throughout. AR_{it} measures the return an asset earns in excess of what its estimated risk exposure predicts on that day.

The cumulative abnormal return for asset i over event window $[\tau_1, \tau_2]$ is:

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{it}. \quad (4.4)$$

I aggregate individual CARs to the portfolio level in two ways. The equal-weighted CAR gives each asset the same weight:

$$\overline{CAR}_g^{EW}(\tau_1, \tau_2) = \frac{1}{N_g} \sum_{i \in g} CAR_i(\tau_1, \tau_2), \quad (4.5)$$

where $g \in \{\text{green, brown}\}$ and N_g is the number of assets in group g . The value-weighted CAR weights each asset by its pre-event market capitalisation:

$$\overline{CAR}_g^{VW}(\tau_1, \tau_2) = \sum_{i \in g} w_i CAR_i(\tau_1, \tau_2), \quad w_i = \frac{MCAP_{i,-1}}{\sum_{j \in g} MCAP_{j,-1}}, \quad (4.6)$$

where $MCAP_{i,-1}$ is the last available market capitalisation before the event date. Pre-event weights are used so that event-window price changes do not alter the portfolio composition.

Reporting both equal-weighted and value-weighted CARs matters because the two aggregations answer different questions. The equal-weighted CAR reflects the return to the average firm in the group and is informative about the breadth of the market reaction. The value-weighted CAR is closer to the return of an investable, size-weighted portfolio and is more relevant for assessing the aggregate wealth effect. Where the two diverge, the response is concentrated among either large or small firms, which itself carries economic content about which types of green or brown firms drove the market reaction.

4.4 Hypotheses and Inference

Hypothesis 1

The first hypothesis is that the oil shock generates a significant cumulative abnormal return for green firms:

$$H_1 : \overline{CAR}_{\text{green}}(\tau_1, \tau_2) \neq 0. \quad (4.7)$$

The substitution channel predicts a positive green response, while the discount-rate channel predicts the opposite. Because both directions are theoretically plausible, I use two-sided tests throughout. Significance stars on negative CARs indicate significant underperformance. Brown CARs are reported as a comparison group under the same framework. Significance is assessed at the one, five, and ten percent levels.

Hypothesis 2

The second hypothesis is that oil-market volatility explains part of any positive green abnormal return. To test this, I augment each baseline model with an oil-volatility variable:

$$r_{it}^e = \alpha_i + \beta_{M,i} MKT_t + \beta_{S,i} SMB_t + \beta_{H,i} HML_t + \beta_{O,i} OilVol_t + \varepsilon_{it}, \quad (4.8)$$

where $OilVol_t$ is one of three oil-volatility proxies. The primary measure is the CBOE Crude Oil Volatility Index (OVX), which is derived from options on crude oil futures and reflects the market's forward-looking expectation of oil price volatility over the next 30 days. OVX rises sharply during periods of geopolitical stress and supply uncertainty, making it the natural measure for the uncertainty channel through which the shock may affect equity prices.

I construct two alternative measures using the Parkinson range-based estimator ([Parkinson, 1980](#)), applied separately to Brent crude and WTI crude daily high and low prices:

$$\sigma_{P,t}^2 = \frac{1}{4 \ln 2} \left(\ln \frac{H_t}{L_t} \right)^2, \quad (4.9)$$

where H_t and L_t are the daily high and low prices of the relevant benchmark. The Parkinson estimator captures realised intraday volatility rather than implied future volatility and is approximately five times more efficient than the squared close-to-close return under a geometric Brownian motion price process. It provides a backward-looking complement to OVX, and using all three proxies ensures that the findings are not driven by the particular construction of one measure.

The key quantity for H2 is the change in the group CAR when the oil-volatility factor is added:

$$\Delta CAR_g = \overline{CAR}_g^{\text{base}} - \overline{CAR}_g^{\text{oil}}. \quad (4.10)$$

A positive ΔCAR means adding the oil-volatility factor reduces the estimated CAR: part of the group's abnormal return is absorbed by the uncertainty channel once it is controlled for. A negative ΔCAR means the augmented model assigns a higher residual return, indicating that oil volatility was suppressing returns rather than inflating them. The ten-day post-event window serves as the primary window for testing H2 among green firms, since it is the only window in which green firms show a positive and significant baseline CAR.

4.5 Placebo Tests

To assess whether the documented return patterns are specific to the March 2026 shock or could arise from the methodology itself, I draw 100 pseudo-event dates at random from the pre-event period $t \in [-285, -35]$ and apply the same samples, factor models, event windows, beta significance filter, and inference procedure to each. The estimation window is re-centred around each placebo date, ending 35 trading days before it, preserving the same structure as the main analysis.

If the results reflect genuine transmission of the oil shock, placebo CARs should be small in magnitude and centred near zero across all windows. A placebo distribution that reproduces the shape or magnitude of the true-event CAR path would indicate that the main findings capture a recurring seasonal pattern, a trend in sector returns, or a persistent feature of the factor models rather than a response to the specific event. The placebo results are discussed in Section 5.3.

Chapter 5

Results

The results reveal a sharply asymmetric equity market response to the Iran-war shock. Brown firms gain immediately, persistently, and across every specification. Green firms fall sharply in the first week, then show a positive equal-weighted signal in the ten days following the event that is concentrated in smaller stocks and is driven by oil-market uncertainty rather than by a revision of expected revenues. The CBOE Crude Oil Volatility Index (OVX) augmentation identifies this channel directly: once implied oil volatility is controlled for, the positive signal in the value-weighted green portfolio disappears entirely. Results using Brent crude realized volatility and WTI crude realized volatility as alternative proxies are reported in Appendix A and discussed in each subsection below. All t -statistics use Newey–West corrections with four lags; all p -values are two-sided.

5.1 Firm-Level Results

5.1.1 Green Firms

Baseline results

The immediate market reaction to the shock is unambiguously negative for green firms. In the three days centered on the event date, I find that the equal-weighted portfolio loses -2.89% under the CAPM ($t = -5.01$) and -3.42% under the Fama–French three-factor model ($t = -5.37$), both significant at the one-percent level (Table 5.1). The value-weighted portfolio loses -0.96% and -1.07% under the same models, though these estimates do not reach conventional significance thresholds. That the equal-weighted decline survives the more demanding Fama–French three-factor benchmark rules out the possibility that the loss is a mechanical consequence of green firms’ growth-stock characteristics: it reflects a genuine, sector-specific repricing. From the standpoint of the research question, the channel is direct—higher expected oil prices raise input costs for renewable energy manufacturers and redirect investor capital toward fossil-fuel producers whose revenues benefit immediately from the price increase (Kilian,

2009).

The seven-day and eleven-day windows centered on the event show the same direction. I find equal-weighted returns of -2.84% under the CAPM and -3.17% under the Fama–French three-factor model over the seven-day window, and -0.83% and -1.15% respectively over the eleven-day window. Value-weighted estimates in these windows are consistently negative but not statistically significant.

The picture changes in the ten days following the event. I find that the equal-weighted portfolio earns $+4.15\%$ under the CAPM ($t = 2.73$) and $+3.96\%$ under the Fama–French three-factor model ($t = 2.62$), both significant at the one-percent level. This is the only positive evidence of a substitution effect in the data: as the price shock persisted and its long-run implications became clearer to investors, part of the market began pricing in higher relative demand for renewable energy (Nerlinger and Utz, 2022).

The standard approach in asset pricing is to evaluate value-weighted returns, as they reflect what an investor holding a market-cap-proportional portfolio would earn (MacKinlay, 1997). The value-weighted portfolio in the ten-day window earns $+3.03\%$ under the CAPM and $+2.74\%$ under the Fama–French three-factor model, but neither estimate is statistically significant. The value-weighted results show that the positive signal is driven by smaller green firms. Smaller pure-play renewable energy companies—whose revenues are tied directly to clean energy production and who carry no offsetting conventional-energy exposure—are most sensitive to a shift in the relative price of energy. Larger green firms, typically diversified utilities or conglomerates, have capital expenditure programmes that make their valuations more sensitive to the higher discount rates that accompany a geopolitical shock, so they benefit less from the substitution narrative and more from the cost-of-capital headwind. The positive equal-weighted return therefore reflects selective rotation into smaller, purer-play green stocks; it is not a broad-based capital reallocation toward the sector.

By twenty-five days after the event, the value-weighted portfolio is significantly negative at -3.46% under the CAPM ($t = -4.70$) and -4.11% under the Fama–French three-factor model ($t = -5.75$). The equal-weighted portfolio shows positive estimates in the fifteen-day window ($+2.85\%$ and $+1.14\%$) but turns negative by twenty-five days ($+2.13\%$ under the CAPM, -0.33% under the Fama–French three-factor model). This reversal is consistent with investors pricing the sustained cost pressures that a persistently high oil-price environment imposes on green firms: energy-intensive clean technology manufacturing faces rising input costs, and the discount-rate headwind on long-duration growth assets intensifies rather than fades as oil prices remain elevated (Hamilton, 2003).

Oil-volatility-augmented results

The CBOE Crude Oil Volatility Index augmentation reveals why the ten-day equal-weighted green return appeared positive. Adding implied oil volatility as an additional regressor reduces

the estimated equal-weighted green return in the ten-day window from +4.15% to +2.72% under the CAPM and from +3.96% to +3.40% under the Fama–French three-factor model. For the value-weighted portfolio, the reduction is larger: the augmented model produces –2.91% (CAPM) and –3.92% (Fama–French three-factor model) against baseline estimates of +3.03% and +2.74% respectively. The baseline return was inflated by a positive co-movement between implied oil volatility and green stock prices during the event window: as oil-market uncertainty rose, green stocks rose with it—not because their expected revenues were revised upward, but because investors were temporarily treating green firms as a hedge against oil-market uncertainty (Baumeister and Hamilton, 2019). This distinction is central to the research question: the positive ten-day green signal reflects a demand for uncertainty hedges, not a fundamental reassessment of energy-transition economics.

Across the three short windows centered on the event date, the augmented model produces estimates that are more negative than the baseline, meaning implied oil volatility was loading negatively on green stocks in those windows. In the immediate aftermath of the shock, elevated implied volatility signalled a deteriorating macro environment for long-duration growth assets, and controlling for this factor reveals that the pure sector-specific green return was more negative than the baseline suggested.

The Brent crude realized volatility and WTI crude realized volatility results in Appendix A run in the opposite direction: augmented estimates exceed the baseline estimates across windows. During the estimation window, high realized oil volatility was historically associated with weaker green firm returns, producing negative loadings on these measures. When the event combined high realized volatility with the positive ten-day green return, removing the historical negative association raises the residual. This divergence shows that the uncertainty-hedge channel is specific to forward-looking implied volatility—the CBOE Crude Oil Volatility Index captures what investors expect oil prices to do, which is what drives portfolio allocation decisions in real time. Realized oil price ranges capture what has already happened and do not generate the same hedge rotation.

5.1.2 Brown Firms

Baseline results

Brown firms respond to the shock immediately, strongly, and persistently. In the three days centered on the event date, I find that the equal-weighted portfolio earns +3.91% under the CAPM ($t = 9.20$) and +4.94% under the Fama–French three-factor model ($t = 11.08$); the value-weighted portfolio earns +3.78% and +5.18% under the same models, all significant at the one-percent level (Table 5.2). A 4–5% abnormal return over three trading days represents a substantial upward revision in the market value of energy producers, consistent with investors pricing in a direct cash-flow windfall as crude prices rise and transit through the Strait of Hormuz becomes constrained. The near-identical equal-weighted and value-weighted estimates

in the short windows indicate that the positive response is broad-based across the size distribution of brown firms: large integrated oil companies and smaller upstream producers gained comparably in the first week.

Returns accumulate through every subsequent window, model, and weighting scheme. In the ten days following the event, I find equal-weighted returns of +5.44% (CAPM) and +6.66% (Fama–French three-factor model); in the fifteen-day window, +9.73% and +10.06%; and at twenty-five days, +11.82% and +9.91%, all significant at the one-percent level across both weighting schemes. The sustained accumulation over three weeks is consistent with investors continuing to revise upward expected fossil-fuel revenues as the disruption persisted, rather than an initial overreaction followed by reversal ([Hamilton and Herrera, 2004](#)). The equity market treated the shock as a lasting revision to the supply outlook.

Oil-volatility-augmented results

The CBOE Crude Oil Volatility Index absorbs only a small fraction of the brown return. In the three-day window, adding implied oil volatility reduces the equal-weighted return by 0.34 percentage points under the CAPM against a baseline of +3.91%. At twenty-five days, the reduction under the Fama–French three-factor model is 3.60 percentage points against a baseline of +9.91%. Positive and statistically significant returns survive augmentation across virtually every window and specification. This is the supply-windfall channel operating without meaningful attenuation: brown firms gain because higher oil prices raise their expected cash flows, and this channel is largely independent of how uncertain the future oil-price path is ([Baumeister and Hamilton, 2019](#)). The contrast with the green result is direct—for green firms, controlling for implied oil volatility substantially reduces the measured positive return; for brown firms, removing the same factor leaves the return nearly intact.

The Brent crude realized volatility and WTI crude realized volatility results in Appendix A produce augmented estimates that are larger than the baseline estimates throughout. High realized oil volatility was historically associated with weaker brown firm performance during the estimation window, so removing this negative loading raises the residual when the event produces both high realized volatility and strongly positive returns. Across all three volatility proxies, the brown return is attributed to the cash-flow channel.

5.2 ETF-Level Results

The ETF results test whether the firm-level patterns are visible through the pooled vehicles that retail and passive investors use to access the green and brown energy sectors. Two features of ETF pricing shape what I find. First, diversification within each fund averages away idiosyncratic firm-level responses, so a signal concentrated in smaller stocks will be attenuated or absent at the ETF level. Second, during volatility spikes, authorised participants reduce ETF

inventory to meet redemptions, which can push ETF prices below the value of the underlying holdings independently of what those holdings are doing (Ben-David et al., 2018).

5.2.1 Green ETFs

Baseline results

Green ETFs fall sharply around the event and show no positive response at any subsequent horizon (Table 5.3). In the three days centered on the event date, I find that the equal-weighted portfolio loses -4.24% under the CAPM ($t = -19.56$) and -4.55% under the Fama–French three-factor model ($t = -17.19$); the value-weighted portfolio loses -4.28% and -4.69% respectively, with t -statistics above thirty in absolute value. The near-identical equal-weighted and value-weighted estimates across all short windows—unlike the divergence observed in the firm sample—reflect the internal diversification of each fund: holding dozens of green firms averages away the size heterogeneity that drives the gap between weighting schemes at the firm level.

In the ten days following the event, the equal-weighted portfolio shows a marginally significant positive return of $+1.81\%$ under the CAPM ($t = 1.71$); the estimate under the Fama–French three-factor model is $+1.74\%$ and not significant at conventional levels. The value-weighted portfolio remains negative at -0.75% and -0.85% in the same window. The positive signal that I find for smaller individual green firms does not aggregate into a detectable ETF-level effect: retail and passive investors who access the green sector through these funds did not earn a positive return at any horizon. By twenty-five days, I find statistically significant negative value-weighted returns of -2.59% under the CAPM ($t = -4.43$) and -3.37% under the Fama–French three-factor model ($t = -13.16$).

Oil-volatility-augmented results

For green ETFs, the CBOE Crude Oil Volatility Index effect runs in the opposite direction from individual green firms. Across all windows, the augmented model produces estimates that are less negative—and at longer horizons, positive—compared to the baseline, meaning that controlling for implied oil volatility reveals that OVX was loading negatively on green ETF prices during the event. At twenty-five days, the difference between the baseline and augmented equal-weighted estimate reaches 11.12 percentage points under the CAPM and 14.44 percentage points under the Fama–French three-factor model: the augmented model produces estimates of $+11.06\%$ and $+13.15\%$ while the baseline is near zero.

The mechanism follows from the authorised-participant structure documented by Ben-David et al. (2018): during the volatility spike, institutional investors redeemed large green ETFs, and authorised participants sold ETF shares in the secondary market to facilitate those redemptions. This selling was correlated with rising implied oil volatility and drove ETF prices

below the value of the underlying holdings. Removing the implied oil volatility component from the return model reveals that a substantial portion of the ETF-level decline was flow-driven rather than a reflection of the underlying green assets. This creates an exactly opposite pattern from what I find at the individual stock level: for individual green firms, implied oil volatility inflated returns through uncertainty-hedge demand; for green ETFs, it depressed prices through flow-driven selling. Controlling for the CBOE Crude Oil Volatility Index is what separates the two channels.

The Brent crude realized volatility and WTI crude realized volatility results in Appendix A show differences between baseline and augmented estimates close to zero across all windows, confirming that the large implied-volatility effect is specific to the forward-looking measure: authorised participants respond to expected uncertainty, not to realized intraday oil price ranges.

5.2.2 Brown ETFs

Baseline results

Brown ETFs earn large, immediate, and persistent positive abnormal returns, consistent with the firm-level results (Table 5.4). In the three days centered on the event date, I find equal-weighted returns of +5.66% (CAPM, $t = 5.24$) and +6.88% (Fama–French three-factor model, $t = 6.45$), with value-weighted returns of +3.63% and +4.98% respectively, all significant at the one-percent level. By twenty-five days, the equal-weighted return reaches +18.14% (CAPM, $t = 5.08$) and +16.83% (Fama–French three-factor model, $t = 4.57$). Equal-weighted returns exceed value-weighted returns at longer horizons, indicating that smaller, more upstream-focused brown ETFs—holding greater concentrations of pure crude oil producers—gained more than large integrated-energy ETFs whose downstream refining and petrochemical operations dilute the direct supply-windfall.

Oil-volatility-augmented results

The CBOE Crude Oil Volatility Index absorbs a growing share of the brown ETF return at longer horizons. By twenty-five days, the difference between the baseline and augmented equal-weighted estimate is 8.32 percentage points under the CAPM and 12.93 percentage points under the Fama–French three-factor model. Short-horizon returns remain large and statistically significant after augmentation, confirming an immediate supply-windfall component at the ETF level. The growing contribution of implied oil volatility at longer horizons is consistent with the geopolitical uncertainty remaining unresolved in the weeks following the shock: beyond the fundamental cash-flow gain, energy producer valuations continued to be supported by an uncertainty premium as investors priced in the possibility of further disruptions to Middle Eastern crude supply (Baumeister and Hamilton, 2019).

The Brent crude realized volatility and WTI crude realized volatility results in Appendix A

produce augmented estimates that exceed the baseline throughout, consistent with the firm-level brown results. The short-horizon supply-windfall is robust across all three volatility proxies.

5.3 Placebo Tests

I repeat the full event study on 100 pseudo-event dates drawn at random from the pre-event period, holding all samples, models, windows, and corrections fixed. The 5th–95th percentile envelope of the resulting placebo cumulative abnormal return paths serves as an empirical reference: a true-event path that falls within this band is statistically indistinguishable from what would arise by chance on a random trading day.

5.3.1 Firm-Level Placebo

For green firms (Figure 5.5, panels a and b), the value-weighted true-event cumulative abnormal returns fall within the placebo envelope throughout the post-event horizon. At the market-cap-weighted level, the green response is indistinguishable from the distribution of returns that arise on random trading days in the pre-event period. This is fully consistent with the firm-level finding: the positive signal in the equal-weighted portfolio in the ten days following the event is concentrated in smaller firms and is too narrow to stand out from normal return variation at the value-weighted level.

For brown firms (Figure 5.5, panels c and d), the true-event cumulative abnormal returns rise sharply above the placebo envelope within days of the event under both the CAPM and the Fama–French three-factor model. No placebo draw produces a response of comparable magnitude or persistence. The brown supply-windfall is specific to the March 2026 shock and cannot be attributed to seasonal patterns, a pre-existing trend in brown firm valuations, or a structural feature of the return models.

5.3.2 ETF-Level Placebo

For green ETFs (Figure 5.6, panels a and b), the true-event cumulative abnormal returns remain within the placebo band throughout, reflecting high cross-draw variability in a sample of twelve funds. There is no evidence of a systematic effect specific to the March 2026 event at the green ETF level.

For brown ETFs (Figure 5.6, panels c and d), the true-event cumulative abnormal returns exceed the placebo envelope after the event under both models. No placebo draw reproduces the double-digit cumulative returns of the true event.

Taken together, the placebo tests corroborate the main results. The brown supply-windfall is an unambiguous departure from the pre-event return distribution. The green response—positive in the equal-weighted portfolio ten days after the event and driven by implied oil-volatility

hedging—is too narrow at the value-weighted level to separate from pre-event noise, which is consistent with the finding that capital reallocation toward green firms was not a broad-based response to the shock.

Table 5.1: Green Firms: Cumulative Abnormal Returns under Baseline and OVX-Augmented Models

Window		CAPM			FF3		
		Base	OVX	Δ	Base	OVX	Δ
[-1, +1]	EW	-2.89*** (-5.01)	-3.74*** (-5.24)	+0.85	-3.42*** (-5.37)	-4.06*** (-5.38)	+0.65
	VW	-0.96 (-2.51)	-2.71** (-4.18)	+1.75	-1.07 (-2.39)	-3.04** (-4.57)	+1.97
[-3, +3]	EW	-2.84** (-2.35)	-4.60*** (-2.71)	+1.76	-3.17*** (-2.63)	-4.48*** (-2.62)	+1.30
	VW	-0.24 (-0.19)	-4.30*** (-6.59)	+4.06	-0.37 (-0.31)	-4.93*** (-6.41)	+4.56
[-5, +5]	EW	-0.83 (-0.52)	-3.12 (-1.33)	+2.29	-1.15 (-0.69)	-2.67 (-1.08)	+1.52
	VW	1.16 (1.64)	-5.13 (-2.56)	+6.30	0.98 (1.67)	-6.09 (-2.56)	+7.08
[0, +10]	EW	4.15*** (2.73)	2.72 (0.94)	+1.43	3.96*** (2.62)	3.40 (1.17)	+0.56
	VW	3.03 (1.99)	-2.91 (-1.25)	+5.95	2.74 (1.95)	-3.92 (-1.87)	+6.67
[0, +15]	EW	2.85 (1.56)	1.21 (0.31)	+1.64	1.14 (0.63)	0.99 (0.25)	+0.14
	VW	2.07 (0.87)	-6.61 (-2.02)	+8.68	1.56 (0.71)	-8.15* (-2.91)	+9.71
[0, +25]	EW	2.13 (1.04)	-0.98 (-0.16)	+3.11	-0.33 (-0.16)	-1.21 (-0.20)	+0.88
	VW	-3.46** (-4.70)	-17.62 (-2.31)	+14.16	-4.11** (-5.75)	-19.94 (-2.62)	+15.83

Notes: The sample contains 106 green firms. CARs are in percentage points. The *baseline* model (Base) estimates abnormal returns using CAPM or the Fama–French three-factor model (FF3), with no oil volatility factor. The *augmented* model (OVX) adds the CBOE Crude Oil Volatility Index as an additional regressor during the event window, capturing the share of returns driven by oil-market uncertainty. $\Delta\text{CAR} = \text{Base} - \text{OVX}$: a positive Δ means OVX absorbs part of the baseline return; a negative Δ means controlling for oil volatility raises the residual return. EW and VW denote equal-weighted and value-weighted portfolios. Newey–West t -statistics (4 lags) in parentheses. Two-sided significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Stars on negative CARs indicate significant underperformance. Estimation window: $t \in [-285, -35]$ relative to the event date of 2 March 2026.

Table 5.2: Brown Firms: Cumulative Abnormal Returns under Baseline and OVX-Augmented Models

Window		CAPM			FF3		
		Base	OVX	Δ	Base	OVX	Δ
[-1, +1]	EW	3.91*** (9.20)	3.57*** (7.59)	+0.34	4.94*** (11.08)	4.32*** (8.86)	+0.62
	VW	3.78*** (8.25)	3.10*** (3.91)	+0.68	5.18*** (11.22)	4.28*** (9.45)	+0.90
[-3, +3]	EW	4.68*** (6.65)	4.03*** (4.11)	+0.65	5.54*** (7.87)	4.26*** (4.40)	+1.29
	VW	3.58*** (4.72)	2.00 (1.22)	+1.58	4.74*** (6.48)	2.67*** (2.87)	+2.07
[-5, +5]	EW	5.55*** (7.58)	4.67*** (3.74)	+0.88	8.67*** (11.25)	6.55*** (5.72)	+2.12
	VW	4.46*** (7.54)	2.32 (1.00)	+2.14	8.39*** (13.25)	5.21*** (5.23)	+3.17
[0, +10]	EW	5.44*** (5.59)	6.38*** (3.74)	-0.94	6.66*** (6.76)	5.03*** (3.28)	+1.63
	VW	5.48*** (6.11)	5.40* (1.86)	+0.08	7.07*** (7.69)	4.61*** (3.48)	+2.45
[0, +15]	EW	9.73*** (8.82)	11.05*** (4.80)	-1.31	10.06*** (8.93)	7.82*** (4.06)	+2.25
	VW	10.65*** (9.99)	10.36** (2.30)	+0.29	11.66*** (11.09)	8.06*** (4.08)	+3.60
[0, +25]	EW	11.82*** (9.51)	13.87*** (4.06)	-2.05	9.91*** (8.22)	6.31** (2.25)	+3.60
	VW	11.35*** (9.74)	10.63 (1.50)	+0.72	9.84*** (8.80)	3.95 (1.23)	+5.89

Notes: The sample contains 153 brown firms. CARs are in percentage points. The *baseline* model (Base) estimates abnormal returns using CAPM or the Fama–French three-factor model (FF3), with no oil volatility factor. The *augmented* model (OVX) adds the CBOE Crude Oil Volatility Index as an additional regressor during the event window, capturing the share of returns driven by oil-market uncertainty. $\Delta\text{CAR} = \text{Base} - \text{OVX}$: a positive Δ means OVX absorbs part of the baseline return; a negative Δ means controlling for oil volatility raises the residual return. EW and VW denote equal-weighted and value-weighted portfolios. Newey–West t -statistics (4 lags) in parentheses. Two-sided significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Stars on negative CARs indicate significant underperformance. Estimation window: $t \in [-285, -35]$ relative to the event date of 2 March 2026.

Table 5.3: Green ETFs: Cumulative Abnormal Returns under Baseline and OVX-Augmented Models

Window		CAPM			FF3		
		Base	OVX	Δ	Base	OVX	Δ
[-1, +1]	EW	-4.24*** (-19.56)	-3.85*** (-6.25)	-0.39	-4.55*** (-17.19)	-3.85*** (-7.11)	-0.70
	VW	-4.28*** (-33.00)	-3.36*** (-7.15)	-0.92	-4.69*** (-46.50)	-3.74*** (-6.40)	-0.95
[-3, +3]	EW	-6.63*** (-7.53)	-5.65*** (-4.32)	-0.98	-6.81*** (-7.60)	-5.11*** (-4.31)	-1.70
	VW	-6.02*** (-5.43)	-3.88*** (-8.11)	-2.15	-6.30*** (-5.90)	-4.07*** (-7.49)	-2.22
[-5, +5]	EW	-4.11*** (-4.42)	-2.02 (-0.93)	-2.09	-4.40*** (-4.67)	-1.12 (-0.53)	-3.28
	VW	-4.41*** (-5.51)	-0.59 (-0.36)	-3.82	-5.15*** (-8.38)	-1.18 (-0.51)	-3.97
[0, +10]	EW	1.81* (1.71)	6.77* (1.72)	-4.96	1.74 (1.64)	8.21** (2.07)	-6.47
	VW	-0.75 (-0.45)	5.27 (0.90)	-6.03	-0.85 (-0.50)	5.42 (0.83)	-6.27
[0, +15]	EW	1.38 (1.35)	8.40 (1.56)	-7.02	0.50 (0.51)	9.56* (1.87)	-9.06
	VW	-1.24 (-0.90)	7.43 (1.01)	-8.67	-1.90 (-1.60)	7.00 (0.90)	-8.89
[0, +25]	EW	-0.05 (-0.07)	11.06 (1.35)	-11.12	-1.29 (-1.56)	13.15* (1.70)	-14.44
	VW	-2.59*** (-4.43)	11.47 (1.15)	-14.06	-3.37*** (-13.16)	11.45 (1.08)	-14.82

Notes: The sample contains 12 green ETFs. CARs are in percentage points. The *baseline* model (Base) estimates abnormal returns using CAPM or the Fama–French three-factor model (FF3), with no oil volatility factor. The *augmented* model (OVX) adds the CBOE Crude Oil Volatility Index as an additional regressor during the event window, capturing the share of returns driven by oil-market uncertainty. $\Delta\text{CAR} = \text{Base} - \text{OVX}$: a positive Δ means OVX absorbs part of the baseline return; a negative Δ means controlling for oil volatility raises the residual return. EW and VW denote equal-weighted and value-weighted portfolios. Newey–West t -statistics (4 lags) in parentheses. Two-sided significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Stars on negative CARs indicate significant underperformance. Estimation window: $t \in [-285, -35]$ relative to the event date of 2 March 2026.

Table 5.4: Brown ETFs: Cumulative Abnormal Returns under Baseline and OVX-Augmented Models

Window		CAPM			FF3		
		Base	OVX	Δ	Base	OVX	Δ
[-1, +1]	EW	5.66*** (5.24)	4.33*** (4.34)	+1.34	6.88*** (6.45)	5.34*** (4.64)	+1.55
	VW	3.63*** (10.07)	1.65*** (5.20)	+1.98	4.98*** (10.46)	3.48*** (7.18)	+1.50
[-3, +3]	EW	6.57*** (4.01)	3.46** (2.30)	+3.11	7.57*** (4.69)	4.01** (2.17)	+3.57
	VW	3.59*** (7.74)	-1.03 (-1.40)	+4.63	4.72*** (9.66)	1.24** (2.17)	+3.47
[-5, +5]	EW	8.50*** (4.04)	3.81 (1.55)	+4.69	11.90*** (5.94)	6.25*** (2.77)	+5.65
	VW	4.48*** (8.53)	-2.73** (-2.26)	+7.21	8.28*** (10.31)	2.78*** (3.42)	+5.50
[0, +10]	EW	11.11*** (3.77)	7.64** (2.28)	+3.47	12.43*** (4.29)	6.89** (2.20)	+5.54
	VW	5.82*** (7.23)	-0.97 (-0.66)	+6.79	7.36*** (8.03)	1.94** (2.07)	+5.42
[0, +15]	EW	15.06*** (5.93)	9.91** (2.37)	+5.15	15.93*** (6.18)	7.82*** (2.66)	+8.11
	VW	10.02*** (9.70)	0.11 (0.05)	+9.91	10.99*** (9.80)	3.08*** (2.74)	+7.91
[0, +25]	EW	18.14*** (5.08)	9.82 (1.48)	+8.32	16.83*** (4.57)	3.90 (0.88)	+12.93
	VW	10.96*** (8.27)	-5.13** (-2.26)	+16.08	9.49*** (7.76)	-3.24** (-2.43)	+12.72

Notes: The sample contains 17 brown ETFs. CARs are in percentage points. The *baseline* model (Base) estimates abnormal returns using CAPM or the Fama–French three-factor model (FF3), with no oil volatility factor. The *augmented* model (OVX) adds the CBOE Crude Oil Volatility Index as an additional regressor during the event window, capturing the share of returns driven by oil-market uncertainty. $\Delta\text{CAR} = \text{Base} - \text{OVX}$: a positive Δ means OVX absorbs part of the baseline return; a negative Δ means controlling for oil volatility raises the residual return. EW and VW denote equal-weighted and value-weighted portfolios. Newey–West t -statistics (4 lags) in parentheses. Two-sided significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Stars on negative CARs indicate significant under-performance. Estimation window: $t \in [-285, -35]$ relative to the event date of 2 March 2026.

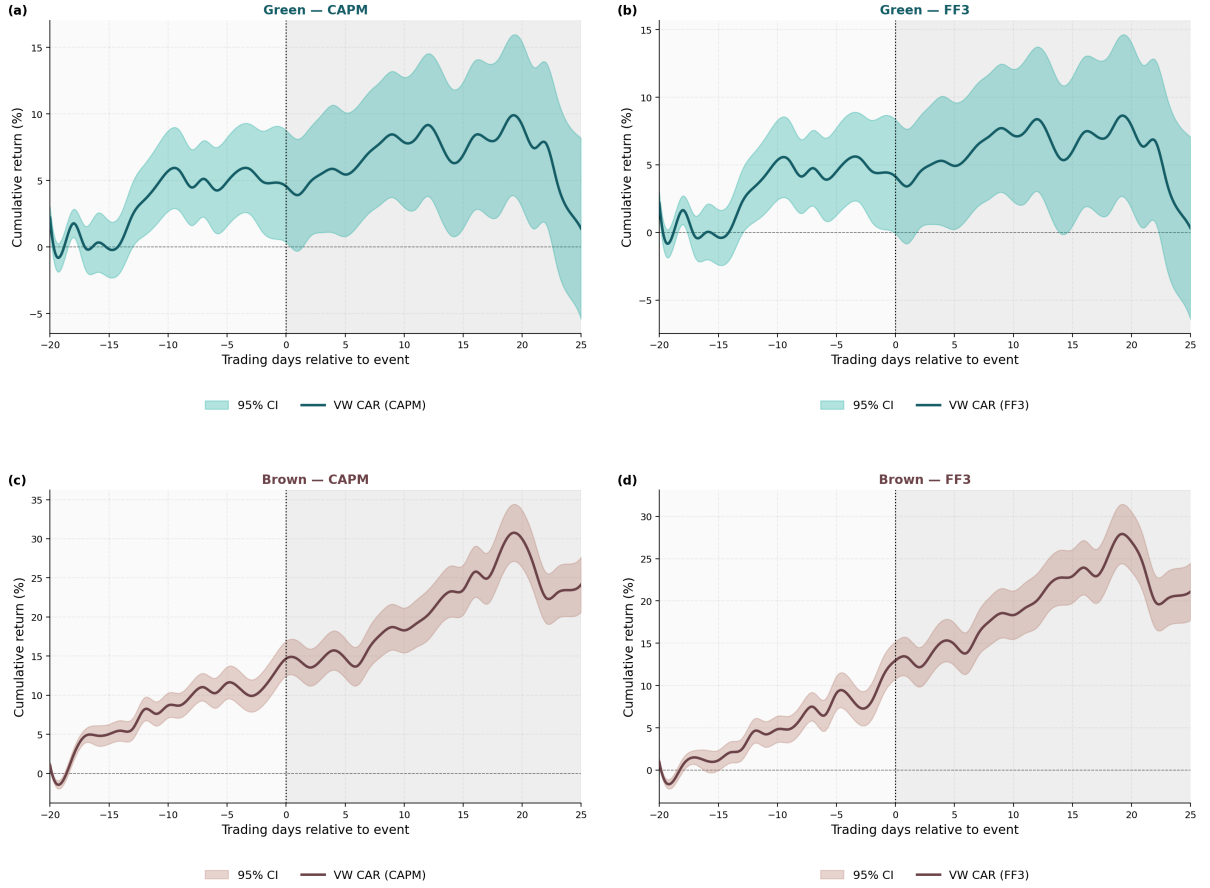


Figure 5.1: Baseline cumulative abnormal returns: firms

Notes: The figure plots value-weighted (VW) CARs (in %) over the window $t \in [-20, +25]$ relative to the event date of 2 March 2026 (vertical dotted line) for the baseline CAPM and FF3 models. The shaded band is the pointwise 95% confidence interval based on Newey–West standard errors (4 lags). Subplots (a) and (b) show green firms ($N = 106$) under CAPM and FF3; subplots (c) and (d) show brown firms ($N = 153$). The estimation window is $t \in [-285, -35]$.

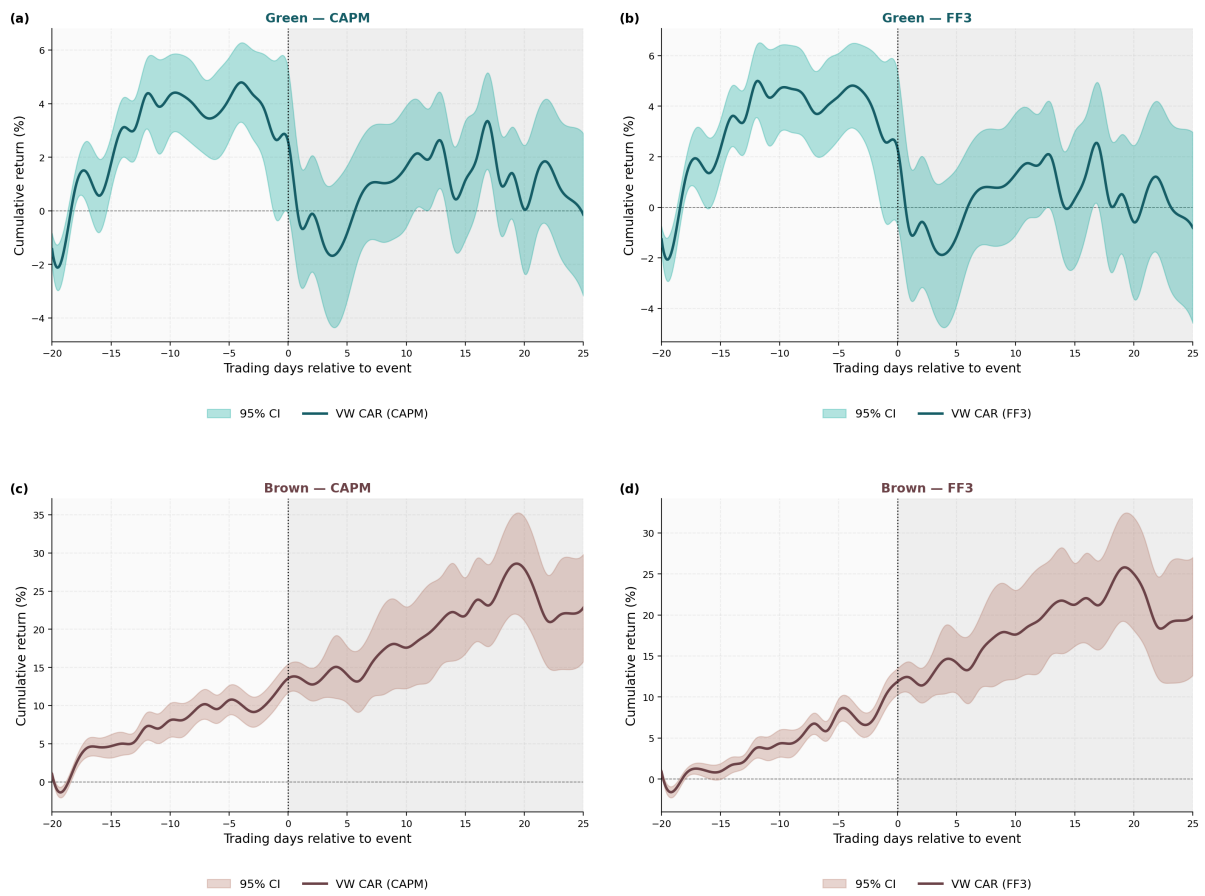


Figure 5.2: Baseline cumulative abnormal returns: ETFs

Notes: As Figure 5.1, for the ETF sample. Subplots (a) and (b) show green ETFs ($N = 12$) under CAPM and FF3; subplots (c) and (d) show brown ETFs ($N = 17$).

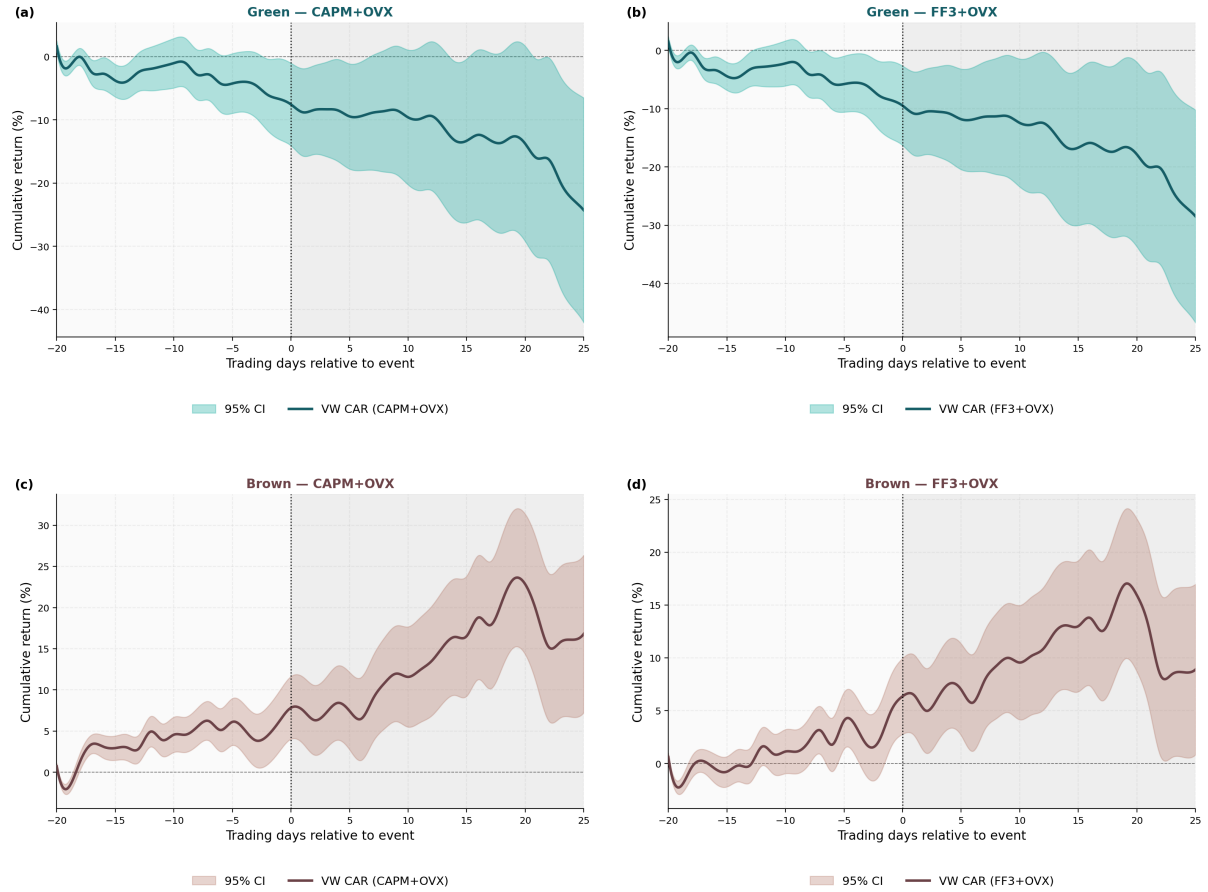


Figure 5.3: OVX-augmented cumulative abnormal returns: firms

Notes: The figure plots value-weighted (VW) CARs (in %) from the OVX-augmented model over $t \in [-20, +25]$ relative to the event date of 2 March 2026 (vertical dotted line). The augmented model adds the CBOE crude oil volatility index (OVX) as an additional factor alongside the market return (CAPM+OVX) or the three Fama–French factors (FF3+OVX). The shaded band is the pointwise 95% confidence interval based on Newey–West standard errors (4 lags). Subplots (a) and (b) show green firms ($N = 106$); subplots (c) and (d) show brown firms ($N = 153$). The estimation window is $t \in [-285, -35]$.

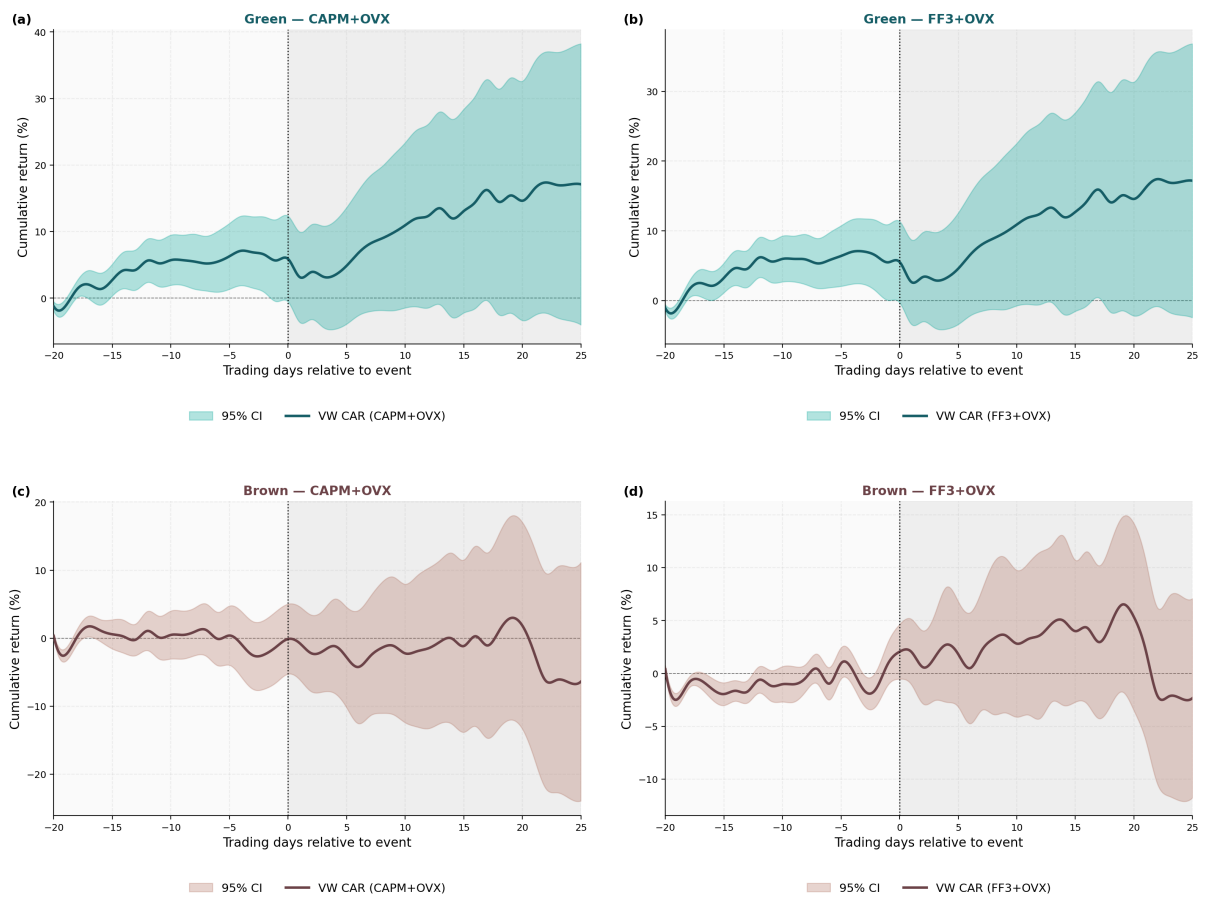


Figure 5.4: OVX-augmented cumulative abnormal returns: ETFs

Notes: As Figure 5.3, for the ETF sample. Subplots (a) and (b) show green ETFs ($N = 12$); subplots (c) and (d) show brown ETFs ($N = 17$).

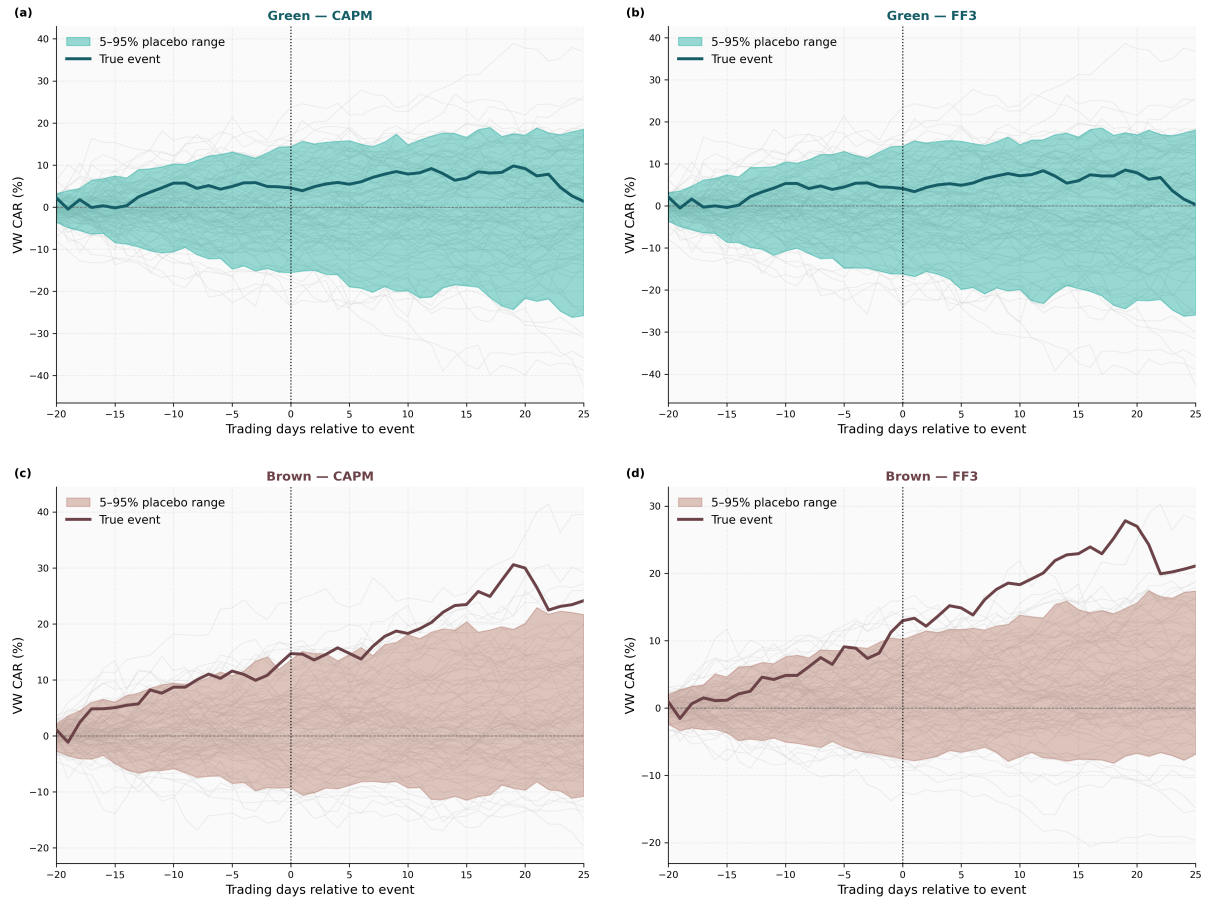


Figure 5.5: Placebo test: Firms

Notes: The figure shows value-weighted (VW) cumulative abnormal returns (CARs, in %) for the true event of 2 March 2026 (dark line) against the 5th–95th percentile envelope (shaded band) from 100 placebo events drawn at random from the estimation period $t \in [-285, -35]$. Subplots (a) and (b) report green firms under CAPM and FF3, respectively; subplots (c) and (d) report brown firms. A true-event CAR outside the placebo band indicates an effect unlikely to arise by chance.

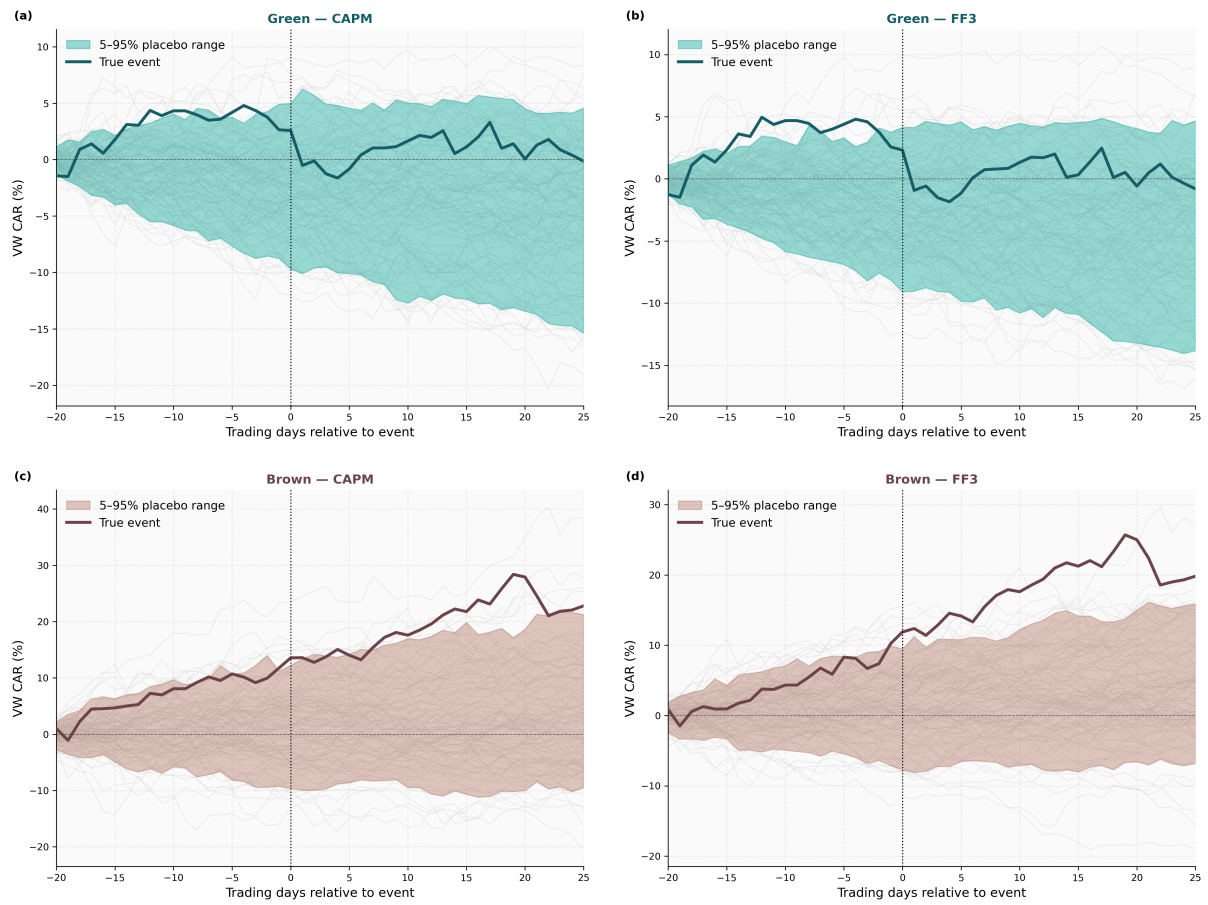


Figure 5.6: Placebo test: ETFs

Notes: As Figure 5.5, for the ETF sample ($N_{\text{green}} = 12$, $N_{\text{brown}} = 17$). Subplots (a) and (b) report green ETFs under CAPM and FF3; subplots (c) and (d) report brown ETFs.

Chapter 6

Conclusion and Limitations

I find a sharply asymmetric equity market response to the Iran-war oil shock of 2 March 2026. Brown firms earn large positive abnormal returns immediately after the shock, and these hold across every window, model, and weighting scheme through the full twenty-five-day horizon. Green firms earn significant negative abnormal returns in the days around the event. In the ten days following the shock, the equal-weighted green portfolio turns positive, but the effect is concentrated in smaller pure-play firms: value-weighted green returns are insignificant at ten days and significantly negative at twenty-five days. Once I control for oil-market uncertainty using the CBOE Crude Oil Volatility Index, the positive ten-day equal-weighted green return shrinks toward zero. I find that the uncertainty channel operates through forward-looking implied volatility; the supply-windfall channel dominates brown returns regardless of the volatility proxy used.

My results contrast with the evidence from the Russia–Ukraine conflict of 2022, when renewable energy stocks earned significant positive abnormal returns ([Nerlinger and Utz, 2022](#); [Umar et al., 2022](#)). Three structural differences explain the divergence. First, the Ukraine shock was primarily a natural gas crisis. Natural gas has closer near-term substitutes in power generation—wind, solar, and other renewables—than crude oil, whose primary end uses in transportation and petrochemicals have no comparable near-term alternative at scale. This made renewables a more direct beneficiary of the 2022 disruption. Second, the geographic routing of the disrupted supply differs substantially. In 2024, China and India together absorbed roughly half of all crude oil and condensate transiting the Strait of Hormuz, while the United States imported less than 3% of its crude consumption through the strait ([Dunn and Barden, 2025](#)). A disruption to Hormuz-routed supply does not generate direct input-cost pressure for U.S.-listed firms, and no substitution incentive follows from a shortage that does not reach U.S. energy markets ([Kilian, 2009](#)). Third, [P’astor et al. \(2022\)](#) show that the large outperformance of green equities during 2020–2021 reflected unexpectedly strong shifts in ESG investor sentiment rather than a permanent rise in expected returns. Once those sentiment shifts were incorporated into valuations, the marginal information content of a further oil price increase for green stock prices was smaller. The Ukraine shock generated exactly this kind of sentiment revision as

European governments accelerated renewable investment commitments. The Iran shock raised oil prices but produced no comparable reassessment of green energy prospects among U.S. equity investors.

A related mechanism concerns the oil price level at the time of the shock. When crude prices are already elevated, the cost competitiveness of renewables is already reflected in equity valuations, and the marginal positive revision from a further price increase is smaller. At high price levels, refiners can economically process a wider range of crude grades and the incentive to expand upstream production intensifies—both effects are cash-flow positive for brown firms but carry no comparable benefit for green firms. [Hamilton \(2003\)](#) establishes that the macroeconomic transmission of oil shocks is state-dependent; the direction and magnitude of the green–brown return differential I document may similarly depend on the price regime in which the shock occurs.

The sign-reversal I find in green returns across horizons is consistent with nonlinear dynamics that a single-regime event study cannot fully capture. Two extensions are suited to this question. A regime-switching event study, drawing on [Hamilton \(1989\)](#), would allow the abnormal return process to differ between high- and low-uncertainty states. [Pérez-Quirós and Timmermann \(2000\)](#) show that small firms exhibit the strongest sensitivity to regime shifts; applied here, their framework predicts that a high-uncertainty state may favour small green firms through the uncertainty-hedge channel, while a supply-windfall state favours brown producers across the size distribution. A threshold regression approach would endogenously identify the oil price level at which the sign of the green equity response changes.

Several limitations apply to my analysis. The sample covers 106 green and 153 brown U.S.-listed firms and may not generalise to markets with a more direct supply-chain link to Middle Eastern crude. I study a single shock, and one event study cannot establish mechanisms that hold across the broader class of geopolitical oil disruptions. The assumption of stable factor loadings over the estimation window is standard ([MacKinlay, 1997](#)) but would be violated if a volatility-regime shift occurs within that window. My findings also carry a clear policy implication: market-based capital reallocation toward green assets is not automatic following an oil supply shock. The positive equal-weighted green response I find is concentrated in smaller pure-play firms and undetectable at the ETF level. Policy measures that reduce the cost of capital for green investment directly—through carbon pricing, green public guarantees, or concessional finance—are likely to be more reliable mechanisms than oil price signals alone.

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Appendix A

Supplementary Tables

Table A.1: Green Firms: Cumulative Abnormal Returns under Baseline and BVOL-Augmented Models

Window		CAPM			FF3		
		Base	BVOL	Δ	Base	BVOL	Δ
[-1, +1]	EW	-3.18***	-2.69***	-0.50	-3.43***	-2.91***	-0.52
	VW	-1.40*	-2.43*	1.03	-1.35	-2.31*	0.95
[-3, +3]	EW	-3.50***	-2.98**	-0.52	-3.45***	-2.86**	-0.58
	VW	-1.08	-2.49**	1.42	-1.06	-2.40**	1.34
[-5, +5]	EW	-2.36	4.71*	-7.07	-1.90	5.31*	-7.21
	VW	-0.20	-13.14	12.94	0.06	-12.16	12.22
[0, +10]	EW	2.62*	11.32***	-8.69	2.93**	11.82***	-8.88
	VW	1.60	-14.59	16.19	1.48	-13.88	15.36
[0, +15]	EW	0.81	12.56***	-11.75	-0.18	12.07***	-12.25
	VW	-0.01	-21.50	21.48	-0.36	-20.73	20.37
[0, +25]	EW	-2.51	11.12**	-13.64	-3.84**	10.51**	-14.35
	VW	-10.53*	-30.81	20.28	-10.78*	-32.79	22.01

Notes: The sample contains 106 green firms. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. BVOL = same model augmented with the Parkinson (1980) range-based realised volatility of Brent crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.2: Brown Firms: Cumulative Abnormal Returns under Baseline and BVOL-Augmented Models

Window		CAPM			FF3		
		Base	BVOL	Δ	Base	BVOL	Δ
[-1, +1]	EW	3.92***	4.58***	-0.65	5.37***	5.55***	-0.18
	VW	4.01***	4.71***	-0.69	5.67***	5.90***	-0.23
[-3, +3]	EW	4.94***	5.66***	-0.73	6.03***	6.19***	-0.17
	VW	4.00***	4.79***	-0.79	5.17***	5.39***	-0.22
[-5, +5]	EW	6.00***	15.83***	-9.82	10.20***	14.61***	-4.41
	VW	5.14***	15.10***	-9.96	9.58***	14.24***	-4.66
[0, +10]	EW	5.71***	18.37***	-12.67	7.10***	13.18***	-6.07
	VW	6.20***	18.83***	-12.63	7.59***	13.74***	-6.15
[0, +15]	EW	10.19***	26.87***	-16.68	10.51***	18.51***	-8.00
	VW	11.74***	28.40***	-16.66	12.44***	20.57***	-8.13
[0, +25]	EW	2.56**	22.13***	-19.57	0.27	9.58***	-9.31
	VW	2.23*	21.57***	-19.34	0.26	9.54**	-9.28

Notes: The sample contains 153 brown firms. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. BVOL = same model augmented with the Parkinson (1980) range-based realised volatility of Brent crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.3: Green ETFs: Cumulative Abnormal Returns under Baseline and BVOL-Augmented Models

Window		CAPM			FF3		
		Base	BVOL	Δ	Base	BVOL	Δ
[-1, +1]	EW	-4.22***	-4.25***	0.03	-4.44***	-4.35***	-0.09
	VW	-4.28***	-4.33***	0.05	-4.52***	-4.53***	0.01
[-3, +3]	EW	-6.67***	-6.83***	0.16	-6.72***	-6.73***	0.00
	VW	-6.05*	-6.19*	0.13	-6.18**	-6.26*	0.08
[-5, +5]	EW	-4.14***	-3.69**	-0.45	-4.22***	-2.52	-1.71
	VW	-4.44*	-4.54*	0.10	-4.84**	-4.38*	-0.46
[0, +10]	EW	1.81	2.49	-0.68	1.89	4.12*	-2.24
	VW	-0.71	-0.77	0.06	-0.78	-0.13	-0.64
[0, +15]	EW	1.38	2.23	-0.85	0.85	3.77	-2.91
	VW	-1.30	-1.43	0.12	-1.63	-0.81	-0.82
[0, +25]	EW	1.49	2.04	-0.54	0.92	4.03	-3.11
	VW	-0.02	-0.43	0.41	-0.13	0.58	-0.70

Notes: The sample contains 12 green ETFs. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. BVOL = same model augmented with the Parkinson (1980) range-based realised volatility of Brent crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.4: Brown ETFs: Cumulative Abnormal Returns under Baseline and BVOL-Augmented Models

Window		CAPM			FF3		
		Base	BVOL	Δ	Base	BVOL	Δ
[-1, +1]	EW	5.86***	6.71***	-0.86	7.35***	7.53***	-0.18
	VW	3.88***	5.08***	-1.20	5.46***	5.44***	0.02
[-3, +3]	EW	6.91***	7.90***	-1.00	7.97***	8.07***	-0.10
	VW	4.05***	5.61***	-1.56	5.17***	4.97***	0.20
[-5, +5]	EW	9.04***	21.00***	-11.95	13.06***	17.51***	-4.45
	VW	5.21***	20.89***	-15.68	9.46***	11.44***	-1.98
[0, +10]	EW	11.70***	26.84***	-15.14	12.96***	18.98***	-6.02
	VW	6.61***	26.32***	-19.71	7.94***	10.79**	-2.86
[0, +15]	EW	15.94***	35.83***	-19.90	16.57***	24.45***	-7.88
	VW	11.18***	37.24***	-26.06	11.85***	15.63**	-3.79
[0, +25]	EW	9.55***	32.93***	-23.38	7.77**	16.74***	-8.96
	VW	2.92*	33.98***	-31.05	1.06	4.92	-3.86

Notes: The sample contains 17 brown ETFs. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. BVOL = same model augmented with the Parkinson (1980) range-based realised volatility of Brent crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.5: Green Firms: Cumulative Abnormal Returns under Baseline and WTIVOL-Augmented Models

Window		CAPM			FF3		
		Base	WTIVOL	Δ	Base	WTIVOL	Δ
[-1, +1]	EW	-3.18***	-2.67***	-0.51	-3.43***	-2.91***	-0.52
	VW	-1.40*	-2.54*	1.14	-1.35	-1.15	-0.20
[-3, +3]	EW	-3.50***	-2.89**	-0.61	-3.45***	-2.82**	-0.63
	VW	-1.08	-2.88**	1.81	-1.06	-0.62	-0.44
[-5, +5]	EW	-2.36	4.94*	-7.29	-1.90	5.31*	-7.21
	VW	-0.20	-13.97	13.76	0.06	0.72	-0.67
[0, +10]	EW	2.62*	11.86***	-9.24	2.93**	12.09***	-9.16
	VW	1.60	-15.96	17.56	1.48	2.06	-0.59
[0, +15]	EW	0.81	12.04***	-11.23	-0.18	11.07***	-11.25
	VW	-0.01	-21.18	21.16	-0.36	0.49	-0.85
[0, +25]	EW	-2.51	11.93**	-14.44	-3.84**	10.82**	-14.66
	VW	-10.53*	-33.24	22.71	-10.78*	-9.03*	-1.75

Notes: The sample contains 106 green firms. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. WTIVOL = same model augmented with the Parkinson (1980) range-based realised volatility of WTI crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.6: Brown Firms: Cumulative Abnormal Returns under Baseline and WTIVOL-Augmented Models

Window		CAPM			FF3		
		Base	WTIVOL	Δ	Base	WTIVOL	Δ
[-1, +1]	EW	3.92***	4.79***	-0.87	5.37***	5.66***	-0.29
	VW	4.01***	4.81***	-0.80	5.67***	5.89***	-0.23
[-3, +3]	EW	4.94***	6.14***	-1.20	6.03***	6.43***	-0.41
	VW	4.00***	5.09***	-1.09	5.17***	5.44***	-0.27
[-5, +5]	EW	6.00***	17.78***	-11.77	10.20***	15.41***	-5.20
	VW	5.14***	16.26***	-11.12	9.58***	14.28***	-4.70
[0, +10]	EW	5.71***	20.91***	-15.20	7.10***	14.13***	-7.02
	VW	6.20***	20.61***	-14.41	7.59***	13.97***	-6.38
[0, +15]	EW	10.19***	28.25***	-18.06	10.51***	18.78***	-8.27
	VW	11.74***	28.90***	-17.16	12.44***	20.02***	-7.58
[0, +25]	EW	2.56**	26.10***	-23.54	0.27	10.92***	-10.65
	VW	2.23*	24.53***	-22.30	0.26	9.94*	-9.68

Notes: The sample contains 153 brown firms. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. WTIVOL = same model augmented with the Parkinson (1980) range-based realised volatility of WTI crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.7: Green ETFs: Cumulative Abnormal Returns under Baseline and WTIVOL-Augmented Models

Window		CAPM			FF3		
		Base	WTIVOL	Δ	Base	WTIVOL	Δ
[-1, +1]	EW	-4.22***	-4.25***	0.03	-4.44***	-4.48***	0.04
	VW	-4.28***	-4.33***	0.05	-4.52***	-4.59***	0.07
[-3, +3]	EW	-6.67***	-6.81***	0.13	-6.72***	-6.87***	0.15
	VW	-6.05*	-6.18*	0.13	-6.18**	-6.35*	0.17
[-5, +5]	EW	-4.14***	-3.59**	-0.55	-4.22***	-3.69**	-0.53
	VW	-4.44*	-4.52*	0.09	-4.84**	-4.98**	0.14
[0, +10]	EW	1.81	2.64	-0.83	1.89	2.70	-0.81
	VW	-0.71	-0.75	0.05	-0.78	-0.88	0.10
[0, +15]	EW	1.38	2.28	-0.90	0.85	1.72	-0.86
	VW	-1.30	-1.43	0.12	-1.63	-1.83	0.20
[0, +25]	EW	1.49	2.31	-0.81	0.92	1.68	-0.76
	VW	-0.02	-0.39	0.37	-0.13	-0.65	0.52

Notes: The sample contains 12 green ETFs. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. WTIVOL = same model augmented with the Parkinson (1980) range-based realised volatility of WTI crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.

Table A.8: Brown ETFs: Cumulative Abnormal Returns under Baseline and WTIVOL-Augmented Models

Window		CAPM			FF3		
		Base	WTIVOL	Δ	Base	WTIVOL	Δ
[-1, +1]	EW	5.86***	6.77***	-0.92	7.35***	7.60***	-0.25
	VW	3.88***	5.30***	-1.42	5.46***	5.46***	-0.00
[-3, +3]	EW	6.91***	8.17***	-1.26	7.97***	8.25***	-0.28
	VW	4.05***	6.20***	-2.15	5.17***	5.04***	0.13
[-5, +5]	EW	9.04***	21.61***	-12.57	13.06***	18.32***	-5.26
	VW	5.21***	23.21***	-17.99	9.46***	11.87**	-2.41
[0, +10]	EW	11.70***	27.98***	-16.28	12.96***	20.12***	-7.16
	VW	6.61***	29.70***	-23.10	7.94***	11.45**	-3.51
[0, +15]	EW	15.94***	35.32***	-19.39	16.57***	25.08***	-8.51
	VW	11.18***	38.87***	-27.69	11.85***	15.97**	-4.12
[0, +25]	EW	9.55***	34.97***	-25.42	7.77**	18.78***	-11.01
	VW	2.92*	39.73***	-36.81	1.06	6.00	-4.95

Notes: The sample contains 17 brown ETFs. Each cell reports the mean CAR (in %). Base = expected return from CAPM or FF3 estimated over $t \in [-285, -35]$. WTIVOL = same model augmented with the Parkinson (1980) range-based realised volatility of WTI crude. Δ = Base CAR – Augmented CAR; a positive value indicates the augmented model absorbs part of the baseline abnormal return. EW = equal-weighted; VW = value-weighted.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$ (two-sided). Stars on negative CARs indicate significant underperformance.